NEUROSCIENCE BLOCK:
Please go over the following points and keep them in mind because I have been in a situation like yours and I know how questions systemically flow. Discuss each question of those in your group and come up with a clear cut answer.

In regards to the divisions of the exam: It will be MCQ, SAQ & OSPE:
For MCQ, it is going to be mostly (70%) Physiology. For SAQ, it will be mostly (70%) Anatomy. For OSPE, it will be mostly Anatomy and Histology (80%)
As far as the Questions and Notes are concerned, they are as follows:

1. You need to know the difference between action and graded potentials.
2. What will happen in case we block the Potassium leak channel? (MCQ)
3. What is the difference between temporal and spatial summation?
4. Does the Na/K+ channel contribute to the action potential?
5. What activates the Sodium channels?
6. How different is the potential in presynaptic Vs. postsynaptic nerves?
7. Where does the graded potential happen in the postsynaptic nerve?
8. How can you determine the axon in a nerve fiber?
9. Release of transmitters into synaptic cleft is dependent upon _ _ _ _ _ ?
10. What is the difference between Ionotropic Vs. metabotropic receptors?
11. What determines whether a neurotransmitter is excitatory or inhibitory?
12. Keep in mind that, generally speaking, the ionotropic channel is usually fast whether excitatory or inhibitory.
13. How many muscle fibers does the alpha motor neuron enervate?
14. IN THE EXAM: If you ant to record EMG, the amplitude of the wave in the big motor unit will be higher than in a small motor unit, why?
15. The Reflex Arc: know it in details (SAQ) … what it consists of + its mechanism.
16. What is the main difference between Muscle spindles and golgi spindles? (EXAM)
17. IN THE EXAM: there will be a question about the difference between intrafusal and extrafusal fibers. (Know them by heart)
18. What is the role of tonically active muscle spindles?
19. What innervates the intrafusal fibers?
20. Which is bigger, alpha or gamma fibers, and how important is that?
21. IN THE EXAM: you will be asked about Golgi Tendon Organ (SAQ).
22. IN THE EXAM: you will be asked about the classification of sensory receptors either in MCQs or SAQs.
23. Why is it difficult to localize pain on the viscera?
24. What are CPGs?
25. What is Hering-Breuer reflex?
26. IN THE EXAM: you will be asked about the classes of movement (voluntary – Reflexes – Rhythmic motor & more complex patterns) you must know these in addition to some examples of each. (EXAM)
27. IN THE EXAM: What is the difference between the Hierarchical and Parallel organization of motor control? (EXAM)
28. What is the difference between medial and lateral brainstem pathways? (EXAM)
29. Which matter (white/gray) in the cerebrum has a higher rate of blood flow?

30. IN THE EXAM: you will be asked about the circle of Willis, what will happen in case we obstruct the blood flow in one of the carotid artery/ in both?
31. What is the most important factor in the regulation of blood flow to the brain?
32. Does high PCO2 means low O2, why?
33. What is the role of perfusion pressure? (EXAM)
34. Explain Cushing Reflex. (EXAM)
35. IN THE EXAM: For the brain Cortexes, you MUST know them all (that is the NAMES) + The Function(s) of Each + subdivision of each cortex + The Blood Supply of Each + What will happen in case it is damaged. (100% in the EXAM)
36. IN THE EXAM: you will be asked IN DETAILS about the data processing in both loops the Direct & the Indirect of the Basal Ganglia. (100% EXAM)
37. IN THE EXAM, you will be asked about the neurotransmitters present in the Basal Ganglia + the function of each on the direct/indirect pathway whether excitatory or inhibitory.
38. What will low levels of Dopamine lead to?
39. Why do we treat parkinson's disease with L-DOPA rather than Dopamine? (100% EXAM)
40. What is the main function of the RAS (Reticular Activating System) & where is it located?
41. What is the EEG most commonly used for?
42. YOU MUST DO THE FOLLOWING: List the 4 main types of waves – Know when each wave is BEST recorded – What wave increases in case we ask the patient to open/close his eye – The frequency of each wave – When it is normally to be found and when not – How different the result will be if the wave is high frequency, low amplitude Vs. high amplitude low frequency – how the waves appear when awake and during sleeping & finally a photo of each. (It would be AWESOME if you made that in a table) (100% in the EXAM)
43. What is an evoked potential and why do we do it?
44. What are the 2 types of sleep? (keep in mind the maximum waves and the minimum waves recorded during each type of sleep)
45. Define EPILEPSY, SEIZURE, CONVULSION and what does each of those involve?
46. Classify Seizures and Epilepsies?
47. What are the main benefits of the CSF?
48. Keep in mind that most of the CSF secretion is active function not passive as in the lymphatics.
49. What is Active Secretion due to? (EXAM)
50. What is the CSF mainly composed of and how important is that?
51. The difference between semantic Vs. episodic memory.
52. YOU MUST KNOW WHAT NEUROTRANSMITTERS ARE PRESENT IN EACH SEGMENT OF THE BRAIN AS LONG AS IT'S MENTIONED IN YOUR LECTURES; DOMAMINE, GLUTAMINE, HESTAMINE, SEROTONIN, ETC.
53. IN THE EXAM, you will be asked about AMNESIA, the definition in addition to its different types.
54. IN THE EXAM, you will be asked about the Vestibulo-ocular reflex, know it in details.
55. YOU MUST know the differences between the functions of the PRIMARY & SUPPLEMENTARY motor cortex in terms of motor movements.
56. YOU MUST know the homunculus of both the Cerebrum and the Cerebellum and what

will happen in case a segment was defected.
57. What is the most important function of the hypothalamus & the other functions (the 4 Fs)?
58. IN THE EXAM, you will be asked about the different chemicals in the: Supraoptic nucleus, Paraventricular nucleus & Suprachiasmatic nucleus.
59. IN THE EXAM: YOU MUST KNOW ALL THE NUCLEI IN THE THALAMUS, THE MOST COMMON CHEMICAL IN EACH NUCLEUS, THE SUBDIVISION OF THE NUCLEUS IF PRESENT & THE FUNCTION OF THE NUCLEUS.
60. How does myelination speed up the AP transmission? What is salutatory conduction?
61. List one genetic and other distructive myelinating disorder.
63. What are the 2 types of nerve fibers responsible for pain perception? (EXAM)
64. IN THE EXAM, you will be asked about Broca's area, Wernicke's area, Lateral prefrontal cortex and Lateral and ventral temporal cortex. KNOW THE LOCATION AND FUNCTION OF EACH AREA (100% IN THE EXAM)
65. What is Aphasia and what is the most common cause of it? (EXAM)
66. KNOW THE DIFFERENCE between Global Aphasia, Broca's Aphasia, Wernicke's Aphasia, Conduction Aphasia, Dysarthia and Dysphonia.
67. For the cerebellum, you need to know the cells located on the molecular layer, perkinje layer as well as the granular layer.
68. You MUST know the different input and output fibers … what is excitatory and what is inhibitory.
69. You MUST know what causes complex spikes and what causes simple spikes.
70. Vestibulo-Ocular Reflex: and most importantly, the eye rotation in the opposite direction.
71. What is Huntington's Chorea?
72. Know the major differences between Huntington's and Parkinson's disease.
73. Know the difference between Parkinson's disease and Parkinsonism.
74. IN THE EXAM: the main causes of unconsciousness.
75. What is the key sign for localizing the lesion coma?
76. How can you infer the cause of unconsciousness when testing the Pupil, Body Temperature & Respiratory Pattern? (EXAM)
77. What's the difference between slowly adapting Vs. rapidly adapting receptors … Spatial Discrimination Vs. Lateral Inhibition … Specific Vs. Non-specific sensory pathways.
78. KNOW PROPRIOCEPTION IN DETAILS (EXAM)
79. Mention the lesions that lead to INTENSE WEAKFULNESS.
80. What hormones/neurotransmitters are released more during day time or during sleep?
81. What is the difference between REM and Slow-wave sleep?
82. What are the characteristics of each stage of the Non-REM sleep (especially stage 2 and 4)? (100% EXAM)
83. Know the 3 layers of the brain and the spinal cord.
84. How to differentiate between an Axon and a Dendrite?
85. Know where the location of Cauda Aquina is (at which vertebral level) (OSPE)
86. You must know the location of each ventricle, where the CSF is secreted from and what is the rout of CSF flow. (EXAM)
87. What forms the gray matter in the spinal cord and what forms the white matter? (EXAM)
88. IN OSPE, you will be having a model of a cerebrum and you will be asked about the precentral gyrus and the postcentral gyrus, so know where the Central Sulcus lies and how to

differentiate it from other sulki.
89. YOU MUST KNOW IN DETAILS the spinal tracts (ALL OF THEM) the ascending and the descending, the sensory and the motor, the ones responsible for pain and the ones responsible for temperature and the ones responsible for touch.
90. YOU MUST KNOW where each tract starts and at what level does it end … what tracts intersect and what tracts don't … what tracts intersect at the level of the cerebellum, brainstem or the level of spinal cord … if the tract is epsilateral or bilateral … All of this will aid you in understanding the affected areas when there's an injury.
91. Know where the spinal cord starts, Foramen Magnum. (OSPE)
92. At what level do we take a PUNCTURE? (MCQ & OSPE)
93. YOU MUST KNOW how to differentiate between Upper Motor Neuron & Lower Motor Neuron lesions. (SAQ & OSPE)
94. Keep in mind that the C1-C7 spinal nerves come out above their vertebral level, C8: between C7 and T1, and from T1 and bellow they originate bellow their corresponding vertebrae.
95. YOU MUST KNOW that the Dura and Arachnoid extend to S2 vertebral level, while Pia matter extends down to the tip of the coccyx as Filum Terminale (OSPE)
96. YOU MUST KEEP IN MIND THAT blood vessels are present inside the subarachnoid space keep pulsing causing the CSF to reach the brain.
97. YOU MUST KNOW the different nuclei present in the spinal cord, knowing their initiation and termination + their function, e.g. substantia gelatinosa, nucleus proprious, etc. (EXAM- MCQ,SAQ,OSPE) REALLY REALLY IMPORTANT.
98. KNOW THE DIFFERENCE BETWEEN 1st, 2nd & 3rd ORDER NEURONS.
99. IN THE SAQ, a case of Anterior cord Syndrome/Brown-Sequard Syndrome cord came in our exam. Know both IN DETAILS.
100. IN THE SAQ, many questions were asked about the cerebral supply as
101. YOU MUST MEMORIZE ALL LGDs BY HEART. They came in our exam, "copy paste"
102. IN OSPE, an X-ray of the neck was present pointing to a vertebra (either the axis/ the atlas) KNOW THEM FROM NOW.
103. IN OSPE, TWO EEG FIGURES CAME, one pointing an arrow to a wave asking you to label it and the other one was for calculating the velocity of the wave.
104. IN OSPE, A MODEL OF THE BRAINSTEM WITH THE CRANIAL NERVES COMING OUT OF IT ASKING FOR THE FUNCTION OF THE GLOSSOPHARYNGEAL NERVE, WHICH IS THE ONE WAS POINTED.
105. IN OSPE, A MODEL FOR THE CEREBRUM WITH POINTING FOR WENIKI'S AND BROCHA'S AREAS ASKING FOR THE NAMES OF THOSE AREAS IN ADDITION TO THEIR FUNCTION + IT WAS ASKED WHICH ONE, IF DAMAGED, WILL CAUSE MOTOR APHASIA AND WHICH BLOOD VESSEL SUPPLIES THE AREA.
106. IN OSPE A MODEL OF THE CRANIUM WAS PRESENT POINTING FOR THE SIGMOID SINUS, WE WERE ASKED TO NAME IT.
107. IN OSPE, A PICTURE OF THE SPINAL COLUMN WAS PRESENT AND WE WERE ASKED WHAT WILL THE SYMPTOMS OF THE PATIENT BE IF THIS PORTION IS DAMAGED, IT WAS THE LATERAL HORN.
108. IN OSPE, A QUESTION WAS ABOUT HISTOLOGY SHOWING THE

INTERSECTION OF TWO LEVELS AND WE WERE ASKED TO DIFINE WHICH LEVEL IS THE HIGHER ONE.

109. IN OSPE, A SECTION OF THE BRAIN WAS PRESENTED AND WE WERE ASKED TO IDENTIFY THE LABLED STRUCTURE (IT WAS THE INTERNAL CAPSULE, SO KNOW ITS SPECIFIC PLACE).

110. IN OSPE, A HISTOLOGICAL PICTURE OF THE CEREBELLUM POITING FOR THE PERKINJE LAYER TO LABLE IT